Combinatorial Algorithms (Math 482), 2026 Spring — Quiz 4

1:50-2:15pm, Monday, March 30, 2026.

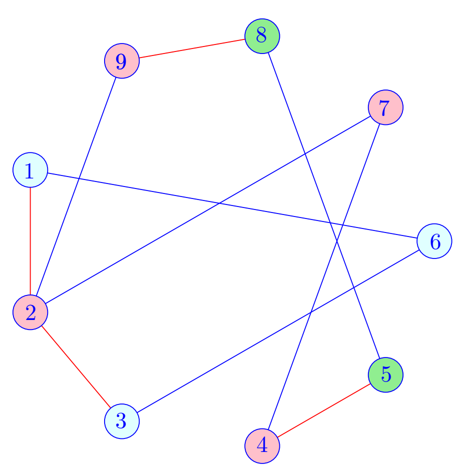
1. **Minimizing Wait Time.** We are given n intervals $(R[1], Q[1]), \ldots, (R[n], Q[n])$, that represent requests for renting a classroom at CSUN. We need to choose an ordering of the requests, and then schedule all requests in this order (back-to-back), so that we allocate $Q[i] - R[i]$ time for request i.

   Dylan suggests that we should sort the requests by $Q[1..n]$ in nondecreasing order. Does this solution minimize the maximum *wait time*? Recall that the wait time of request i is defined as $S[i] - R[i]$ (that is, actual start time minus requested start time).

   Answer. Yes, we know from Problem 2 in Problem Set 7 that this algorithm is optimal.

2. **Classifying Roundworms.** You designed an algorithm that can determine whether m pairwise comparisons between n roundworms is consistent with the hypothesis that all specimens belong to only two subspecies. Run your algorithm on the data below for $n = 9$ specimens, $V = \{1, 2, \ldots, 9\}$, where the following 10 pairs have been compared.
   Same subspecies: (1,6), (2,7), (2,9), (3,6), (4,7), (5,8).
   Different subspecies: (1,2), (2,3), (4,5), (8,9).

   Answer. Yes, the data is consistent with the hypothesis. For example, the two subspecies may be $\{1, 3, 5, 6, 8\}$ and $\{2, 4, 7, 9\}$.



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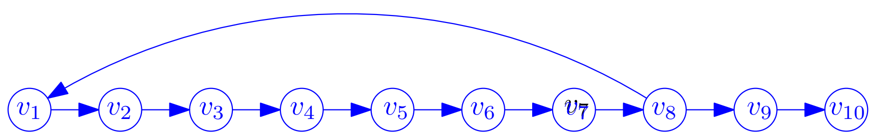
3. **Strongly connected components.** Construct a directed graph $G = (V, E)$ with the following specifications: G has exactly 10 vertices and exactly 3 strongly connected components; there are two special vertices $s, t \in V$ such that G contains directed paths from s to all other vertices, as well as directed paths from all vertices to t.

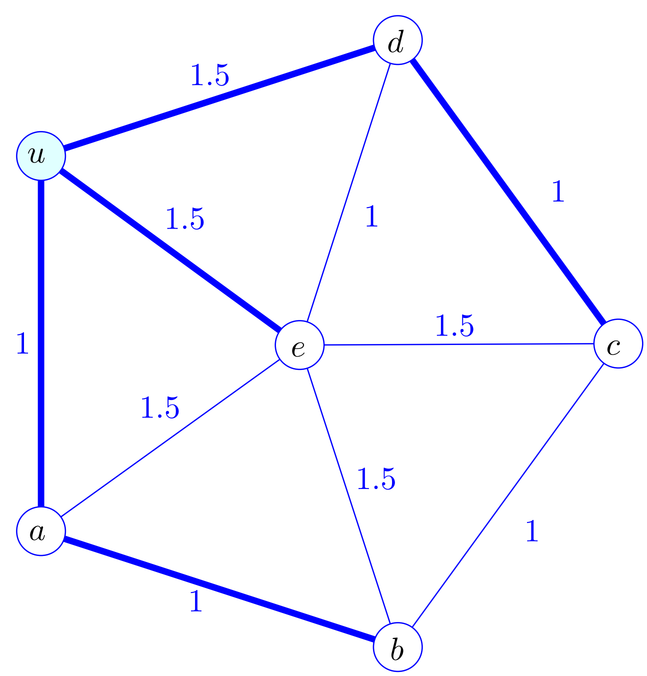
Answer. Consider the graph below with $s = v_1$ and $t = v_{10}$.



The strongly connected components are $\{v_1, v_2, \ldots, v_8\}$, $\{v_9\}$, and $\{v_{10}\}$.

4. **Shortest path trees.** Assign positive weights to the edges of the graph below so that the shortest path tree $T(u)$ rooted at u in <u>not</u> an MST of the graph. (Indicate edge weights and highlight the tree $T(u)$ in the picture. Explain why $T(u)$ is not the MST.)

Answer. With the edge weights indicated below, $T(u)$ consists of the bold edges.



The tree $T(u)$ is <u>not</u> the MST, because its weight is 6, and the 5 edges of unit weight form an MST of weight 5.